UNIT 6 · STANDARD 6.EE.1

Powers and Exponents

Lesson 6-1

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Picture first, then the word, then a plain-language meaning. Say each word out loud.

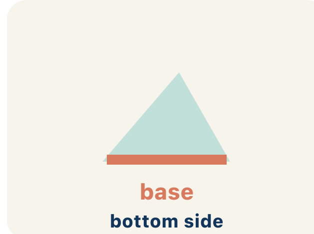
$$5^{3}$$

$5 \times 5 \times 5$

In 2^3, the small 3 means multiply 2 by itself 3 times: $2 \times 2 \times 2 = 8$

Exponent

A small number that tells how many times to multiply the number by itself.



In 5^2, the base is 5 — it is the number being multiplied: $5 \times 5 = 25$

Base

The number that gets multiplied by itself.

$$5^{3}$$

$5 \times 5 \times 5$

$10^3 = 10 \times 10 \times 10 = 1{,}000$ — read as '10 to the third power' or '10 cubed'

Power

A number written with a base and an exponent, like 2^3.

$$3x + 5$$

no equal sign

Evaluate 3^4: write $3 \times 3 \times 3 \times 3$, then multiply step by step: $9 \times 9 = 81$

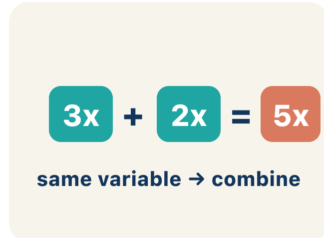
Evaluate

To find the value of an expression.

In 3x, the 3 is the coefficient — if x = 4, then 3x = 3 × 4 = 12

Coefficient

The number in front of a letter, like the 3 in 3x.



4x and 2x are like terms (both have x); 4x and $4x^2$ are NOT like terms (different powers)

Like terms

Terms with the same letter, like 2x and 5x.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can write and evaluate numbers in exponent form using a base and a power.

1️⃣ Notice

You're a sound engineer at a music studio.

2️⃣ Key idea

I can write and evaluate numbers in exponent form using a base and a power.

3️⃣ Words I'll use

Exponent **Base** **Power** **Evaluate** **Coefficient** **Like terms**

🤔 Think About It

- What happens to the volume each time you flip a switch?
- Why is 2^3 equal to 8, not 6?
- How is repeated multiplication different from repeated addition?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

👀 **I do — watch**

Follow each step as your teacher solves it.

Problem: What is the value of 4^3?

A. 64

B. 12

C. 43

D. 7

Step 1 $4^3 = 4 \times 4 \times 4 = 64$.

✅ **Answer:** A. 64

🤝 **We do — together**

Solve this one with your class using the same steps.

Problem: What is the value of 6^2?

A. 36

B. 12

C. 62

D. 8

Step 1 ______________________________

Step 2 ______________________________

Answer: ______________________________

✏️ **You do — your turn**

Now try one on your own. Show every step.

Problem: Which expression shows 5^3 as repeated multiplication?

A. $5 \times 5 \times 5$

B. 5×3

C. $5 + 5 + 5$

D. $3 \times 3 \times 3 \times 3 \times 3$

Show your work:

____

____

____

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

In the sound studio, the volume doubles each time you flip the switch. Why is the volume after 3 flips $2^3 = 8$ and not $2 \times 3 = 6$?

Level 1 support

Start here: The exponent counts how many times we multiply, not what we add.

💡 **Need a hint?**

Sentence starters (optional)

After 3 flips the volume is ____ because we multiply 2 three times.

Después de 3 cambios el volumen es ____ porque multiplicamos 2 tres veces.

It is not 6 because doubling means we ____, not add.

No es 6 porque duplicar significa que ____, no sumar.

Word bank: Exponent | Base | Power | Evaluate

Listen for: *Students connect each switch flip to one more factor of 2 and explain that $2^3 = 2 \times 2 \times 2 = 8$, distinguishing repeated multiplication from repeated addition (which would give 6).*

Level 2

If the volume tripled instead of doubled each flip, how would you write the volume after 3 flips, and how many times louder would that be than the doubling case?

- Tripling gives ____ = ____, which is ____ times the doubling result because ____.
- Changing the base from 2 to 3 matters more than changing the exponent because ____.

EXPLORE

Look at the table: when the exponent goes up by 1 (like from 2^3 to 2^4), what happens to the value, and why?

Level 1 support

Start here: Each time the exponent grows by 1, the value is multiplied by the base one more time.

Need a hint?

Sentence starters (optional)

When the exponent increases by 1, the value is multiplied by ___.

Cuando el exponente aumenta en 1, el valor se multiplica por ___.

This happens because the base is used as a factor ___ more time.

Esto pasa porque la base se usa como factor ___ vez más.

Word bank: **Exponent** **Base** **Power** **Evaluate**

Listen for: *A strong answer states that the value gets multiplied by the base (e.g., $2^3 = 8$ becomes $2^4 = 16$, times 2) and ties it to adding one more factor, using the words base and exponent.*

Level 2

Compare 2^5 and 5^2. Both use the digits 2 and 5, so why are they not equal? Generalize: does the bigger exponent or the bigger base have more effect on the value?

- 2^5 = ___ but 5^2 = ___, so they are not equal because ___.
- In general, a larger ___ has more effect because ___.

CONNECT

Each recording layer doubles the file size, modeled by 4×2^n. Why does the file size grow so fast as you add layers?

Level 1 support

Start here: Doubling repeatedly is exponential growth, so the size explodes after only a few layers.

💡 **Need a hint?**

Sentence starters (optional)

After 5 layers the file is ___ MB because 4×2^5 = ___.

Después de 5 capas el archivo es ___ MB porque 4×2^5 = ___.

It grows quickly because each layer ___ the size.

Crece rápido porque cada capa ___ el tamaño.

Word bank: Power | Base | Exponent | Evaluate

Listen for: *Students evaluate $4 \times 2^5 = 4 \times 32 = 128$ MB and explain that exponential (doubling) growth multiplies the size each step, unlike adding a fixed amount.*

Level 2

Critique this claim: 'After 10 layers the file is just twice as big as after 5 layers.' Is that true? Justify with the powers involved.

- The claim is ___ because 2^{10} compared to 2^5 is ___, not just doubled.
- Exponential growth means going from 5 to 10 layers multiplies the size by ___.

Write About the Math

The Writing Revolution

I can explain my work using the words exponent, base, power, and evaluate.

1. Kernel Sentence

subject + verb

Model: Exponent is a small number that tells how many times to multiply the number by itself.
Exponente es un número pequeño que dice cuántas veces multiplicar el número por sí mismo.

Write a kernel sentence about exponent. Use a subject and a verb.
Escribe una oración base sobre exponente. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Exponent matters in math
Exponente importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Exponent matters in math because ____.
Exponente importa en matemáticas porque ____.

____

but
pero

Exponent matters in math, but ____.
Exponente importa en matemáticas, pero ____.

____

so
entonces

Exponent matters in math, so ____.
Exponente importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about exponent.

Di un hecho verdadero sobre exponent.

Exponent ____.

Question
Pregunta

Ask a question about exponent.

Haz una pregunta sobre exponent.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about exponent.

Muestra entusiasmo sobre exponent.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with exponent.

Dile a un compañero qué hacer con exponent.

First, ____ .

Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

After 3 flips the volume is ____ because we multiply 2 three times.

Después de 3 cambios el volumen es ____ porque multiplicamos 2 tres veces.

It is not 6 because doubling means we ____, not add.

No es 6 porque duplicar significa que ____, no sumar.

When the exponent increases by 1, the value is multiplied by ____.

Cuando el exponente aumenta en 1, el valor se multiplica por ____.

Try It

Solve on your own. Check the answer key when you are done.

1. Which expression equals 81?

A. 3^4

B. 3^3

C. 2^6

D. 4^3

Show your work:

2. What is 5^3?

A. 125

B. 15

C. 25

D. 8

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A bacteria colony doubles every hour. Starting with 1 bacterium, write a power expression for the number after 6 hours. Explain why exponential growth is so much faster than adding the same number each hour.

Sentence starter: After 6 hours there are ____ bacteria because 2^6 = ____. If it grew by adding 2 each hour instead, there would only be ____ bacteria. Exponential growth is faster because ____.

Show your work:

Reflect — Exit Ticket

What is the value of 3^4?

A. 81

B. 12

C. 34

D. 64

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 36 *— $6^2 = 6 \times 6 = 36$.*
2. **You Try (notes):** A. $5 \times 5 \times 5$ *— 5^3 means use 5 as a factor 3 times: $5 \times 5 \times 5$. The exponent tells how many times to multiply, not what to multiply by.*
3. **Try It 1:** A. 3^4 *— $3^4 = 3 \times 3 \times 3 \times 3 = 81$. $3^3 = 27$, $2^6 = 64$, and $4^3 = 64$.*
4. **Try It 2:** A. 125 *— $5^3 = 5 \times 5 \times 5 = 125$.*
5. **Exit Ticket:** A. 81 *— $3^4 = 3 \times 3 \times 3 \times 3 = 81$.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Exponent is a small number that tells how many times to multiply the number by itself.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).